A Domain-Split Mixture-of-Experts Framework for Four-Class Mammography Lesion Detection on CBIS-DDSM

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Abstract. Automated mammography screening must handle two radiologically distinct lesion families—masses and calcifications—that differ markedly in scale, texture, and detection difficulty. Masses appear as irregular soft-tissue opacities spanning tens to hundreds of pixels at standard acquisition resolution, whereas microcalcification clusters occupy only a few pixels once a full-field mammogram is downscaled to a 1280-pixel network input. Training a single detector jointly on all four CBIS-DDSM classes (benign mass, malignant mass, benign calcification, malignant calcification) forces the model to optimise a shared loss over heterogeneous subtypes, potentially suppressing performance on either. We propose a Domain-Split Mixture-of-Experts (DS-MoE) framework that assigns dedicated YOLO11l detectors to mass and calcification subtypes separately and merges their predictions at inference through a class-offset union requiring no routing decision. A Gated MoE variant modulates expert confidence scores using a lightweight image-level router. All experiments use patient-level GroupShuffleSplit partitions to eliminate label leakage, and three random seeds are evaluated to characterise variance. On the 458-image CBIS-DDSM test set, DS-MoE achieves a mean mAP50 of 0.0649 ± 0.0114, a 13.9% improvement over the universal four-class baseline (0.0570 ± 0.0014), with gains of 27.0% and 15.0% on benign and malignant mass classes respectively. Free-Response ROC (FROC) sensitivity at two false positives per image (S@2) is equivalent between DS-MoE and Universal (0.2027 ± 0.0114 vs 0.2021 ± 0.0071), confirming no aggregate clinical sensitivity regression. On held-out INbreast mammograms without fine-tuning, DS-MoE mass FROC S@2 exceeds the universal model by 11.6%. External evaluation on MIAS shows consistent mAP50 improvement (+4.2%), and on CMMD image-level sensitivity improves 77-fold. The calcification expert is identified as the system bottleneck; resolution constraints at standard YOLO input sizes limit its contribution.

Keywords: breast cancer screening · calcification detection · CBIS-DDSM · FROC analysis · mammography lesion detection · mixture of experts · object detection · YOLO

1 Introduction

Breast cancer remains the most frequently diagnosed malignancy in women globally, with approximately 2.3 million new cases reported in 2020 [1]. Mammographic screening has been shown to reduce breast cancer mortality by detecting lesions at earlier, more treatable stages [2], and it remains the primary modality in national screening programmes worldwide. Despite this, a meaningful fraction of breast cancers are missed at initial reading. Retrospective analyses consistently show that missed cancers fall into two distinct visual categories: mass-like opacities that were present but overlooked, and subtle microcalcification clusters that were not flagged [3]. Computer-aided detection (CAD) systems are intended to reduce these miss rates, yet translating benchmark performance into clinical utility has proven harder than anticipated.

One structural reason is that mass and calcification lesions do not share the same detection challenge. A mass occupies a recognisable blob of tissue at multiple scales; its edges interact with surrounding parenchyma in ways that can be captured by standard feature pyramid representations. A microcalcification cluster, by contrast, may span only 5–15 pixels in the original 3000×4000 pixel mammogram. After network downscaling to a 1280-pixel input, those pixels collapse to as few as one or two, below the reliable receptive field of any standard convolutional anchor. This resolution mismatch is not merely a data insufficiency problem; it is a structural mismatch between what the network can represent and what the target looks like after preprocessing.

Most published CAD pipelines address multi-class mammography detection by training a single model on all lesion categories simultaneously [6]. The rationale is straightforward: a shared backbone can reuse low-level features across lesion types, and joint training is simpler to engineer. The cost is that the optimiser must balance loss contributions from classes with very different difficulty profiles. When calcification detection collapses at standard resolution, the gradient signal from calc images provides little useful information to the backbone, and the network effectively specialises on mass detection by default—while still being penalised for calc misses. This implicit bias is invisible in aggregate metrics but becomes apparent when per-class results are examined.

Mixture-of-Experts (MoE) architectures [10] offer a principled alternative: route inputs to specialist subnetworks, each optimised for a well-defined subset of the problem. In sparse variants [11] a hard top-k selection activates only the most relevant experts, allowing model capacity to scale without proportional per-input compute overhead. These ideas have been applied widely in natural language processing but have received limited evaluation in medical object detection, where class heterogeneity is severe and labelled data are scarce.

We investigate whether a simple domain-split MoE—one expert per lesion family, no routing decision, predictions merged by class-offset union—can improve four-class mammography detection on the CBIS-DDSM benchmark [4]. To characterise the value of routing, we also evaluate a Gated MoE variant that uses a lightweight image classifier to modulate expert confidence scores before merging. All models share an identical patient-level data split so that the com-

parison is unconfounded by leakage. We evaluate on three seeds to characterise variance, and assess zero-shot generalisation on INbreast [5], MIAS, and CMMD.

The main contributions of this paper are as follows:

1. We introduce the Domain-Split MoE (DS-MoE) framework for four-class mammography detection, where specialist YOLO11l detectors for mass and calcification domains are trained independently and merged at inference through a parameter-free class-offset union—achieving 13.9% higher mAP50 than a leakage-free universal baseline with mass class gains of 27.0% and 15.0%, while aggregate FROC S@2 remains clinically equivalent.
2. We present, to our knowledge, the first patient-level leakage-free evaluation of a mixture-of-experts mammography detector across three external datasets (INbreast, MIAS, CMMD), showing zero-shot cross-scanner generalisation (+11.6% mass FROC S@2 on INbreast, 77-fold sensitivity gain on CMMD).
3. We diagnose per-class FROC regression in DS-MoE as a consequence of cross-domain false-positive inflation from the calc expert, and quantify the tradeoff between mAP50 gains and per-class sensitivity.
4. We demonstrate through controlled ablation that CLAHE preprocessing is harmful for YOLO-based calcification detection on CBIS-DDSM (−38.6% mAP50), a negative result directly actionable for practitioners building mammography CAD pipelines.

2 Related Work

2.1 Mammography CAD and Lesion Detection

Computer-aided detection in mammography predates deep learning by two decades [3]. Classical pipelines combined Laplacian-of-Gaussian blob detectors for calcification clusters with morphological descriptors for mass shape and rule-based classifiers for BI-RADS assessment. These systems performed reasonably within the institution that developed them but showed substantial degradation across scanner vendors and acquisition protocols, limiting clinical adoption.

The release of CBIS-DDSM [4] and INbreast [5] made large-scale convolutional approaches tractable. Ribli *et al.* [6] applied a Faster R-CNN variant to DDSM images and demonstrated that end-to-end detection networks could approach radiologist sensitivity on held-out cases. Since then, feature pyramid networks, multi-scale attention, and self-supervised pre-training have progressively improved benchmark numbers. A persistent challenge across all these methods is severe class imbalance between benign and malignant subtypes, which biases models toward majority classes even when minority-class recall is clinically more important. Generalisation across acquisition modalities—screen-film to full-field digital—remains non-trivial, reflecting distributional shift in image contrast, noise characteristics, and glandular density.

A critical but often overlooked issue in the literature is data leakage. Many papers split CBIS-DDSM at the image level, inadvertently placing views of the same patient in both train and test partitions. Lee *et al.* [4] explicitly warn

that image-level splits can inflate detection metrics by 30–50% or more, making honest cross-paper comparison impossible. Our work enforces strict patient-level partitioning via `GroupShuffleSplit` throughout all experiments; direct numerical comparison with prior work is therefore not meaningful and is not attempted.

2.2 YOLO-Based Detection in Medical Imaging

The YOLO family of single-stage detectors [7] has gained traction in medical imaging applications due to its inference speed and competitive accuracy. YOLOv8 and YOLO11 [8] use path aggregation network necks with multi-scale prediction heads at strides 8, 16, and 32. The optional P2 head adds a stride-4 output specifically for very small objects, and has shown utility in satellite imagery and digital pathology where the object-to-image area ratio is extremely low. Applied to full-field mammography, YOLO-based detectors face the challenge that calcification clusters occupy a negligible fraction of image area after downscaling, while masses are more amenable to standard anchor-based detection.

Sliced Inference with Hyper Inference (SAHI) [9] addresses small-object detection at inference time by dividing the input into overlapping tiles, running the detector on each tile independently, and merging predictions with non-maximum suppression. This recovers detections lost after global downscaling, at the cost of inference time proportional to tile count. For mammography calcification clusters, SAHI is a natural candidate to address the resolution bottleneck identified in this work; we evaluate tiling as a direction in Section 7 and leave full SAHI integration to future work.

2.3 Mixture-of-Experts for Vision

The MoE framework of Jacobs *et al.* [10] trained specialist networks with a gating network that assigned input samples to the most relevant expert via soft weighting. Shazeer *et al.* [11] introduced sparse top-k gating that scales MoE model capacity without proportional compute cost, enabling the very large MoE transformers now common in NLP. In vision, MoE has been applied mainly to classification and scene understanding tasks. Applying hard routing to object detection is complicated because a single image can contain lesions from multiple classes, so assigning the entire image to one expert risks suppressing detections from the other. Our Domain-Split approach avoids this by running all experts on every image and merging by class partition, making it robust to images that contain both mass and calcification lesions simultaneously.

3 Dataset

3.1 CBIS-DDSM

The Curated Breast Imaging Subset of DDSM (CBIS-DDSM) [4] contains digitised screen-film mammograms with standardised ROI bounding box annota-

Table 1. Dataset partitions used in all experiments. Expert subsets (canonical_mass, canonical_calc) use their own val sets during training; all system-level results report on the shared 458-image canonical_4class test set.

Subset	Classes	Train	Val	Test
canonical_mass	BM, MM	604	26	26
canonical_calc	BC, MC	483	17	18
canonical_4class	BM, MM, BC, MC	2190	455	458
canonical_router	mass / calc	760	162	165

tions for masses and calcification clusters. We work with a four-class lesion detection formulation: *benign_mass* (BM), *malignant_mass* (MM), *benign_calc* (BC), and *malignant_calc* (MC). After cleaning duplicate records and resolving filename collisions, the dataset contains 3,242 usable records. The fixed test partition used in all experiments contains 458 images with 518 ground-truth bounding boxes: BM = 147, MM = 106, BC = 164, MC = 101. Calcification annotations delineate the bounding box of the entire cluster, not individual specks—an important distinction when comparing to external datasets in Section 6.

3.2 Patient-Level Split

All four subsets in Table 1 derive from a single patient-level `GroupShuffleSplit` [12] with 80/10/10 train/val/test ratio applied at the patient identifier level. Every patient identifier appears in exactly one partition, preventing label leakage. The split uses seed 42 as the master partition; all three experimental seeds (42, 1, 123) produce independently trained models but evaluate on the same 458-image test patients, ensuring a fair comparison. With $n = 3$ pairs the minimum achievable Wilcoxon signed-rank p-value is 0.25; we therefore do not claim statistical significance and report mean ± population standard deviation throughout.

4 Methodology

4.1 Problem Formulation

We formulate mammography lesion detection as a four-class object detection problem over full-field mammograms. Given an input image $\mathbf{x} \in \mathbb{R}^{H \times W}$ representing a single mammographic view, the goal is to produce a set of predictions $\mathcal{P} = \{(b_i, c_i, s_i)\}$ where b_i is a bounding box, $c_i \in \{0, 1, 2, 3\}$ is the class index (BM, MM, BC, MC), and s_i is the detection confidence score. The challenge is that the four classes span two fundamentally different detection regimes: mass lesions (classes 0–1) are blob-like objects at centimetre scale, while calcification clusters (classes 2–3) are point-like objects at millimetre scale that become subpixel after standard network downscaling. A single shared model must balance these regimes; our framework addresses this by domain splitting.

4.2 Data Preparation

Each canonical subset is constructed by filtering CBIS-DDSM records for the relevant lesion types and converting bounding box annotations to YOLO format (normalised x_c, y_c, w, h per image). For canonical_mass and canonical_calc, class indices are reassigned to a local two-class scheme: BM/MM $\rightarrow$ 0/1 for mass, and BC/MC $\rightarrow$ 0/1 locally for calc. At inference, a fixed offset of +2 maps calc-expert predictions back to the global four-class index space. Images are converted to 8-bit grayscale and saved as JPEG. No CLAHE preprocessing is applied; Section 5.5 shows that CLAHE is harmful for YOLO-based calcification detection. Expert model selection during training uses the expert's own val partition (26 images for mass, 17 for calc); all final system-level comparisons use the shared 458-image canonical_4class test set.

4.3 Expert Detector Training

Both specialist detectors use YOLO11l [8], a large-scale single-stage detector with a path aggregation network neck and multi-scale prediction heads at strides 8, 16, and 32. The input resolution is set to 1280 pixels for all models to maximise mass lesion visibility while remaining within GPU memory constraints on NVIDIA T4 hardware (batch size 16, AMP enabled).

Training uses the AdamW optimiser with initial learning rate $\eta_0 = 5 \times 10^{-4}$, cosine annealing to $\eta_f = \eta_0 \times 10^{-2}$, weight decay $\lambda = 5 \times 10^{-4}$, and 10-epoch linear warmup. Early stopping monitors validation mAP50 with patience 30 over a maximum of 150 epochs.

Augmentation is domain-adapted: horizontal and vertical flip (probability 0.5 each), colour jitter, and copy-paste ($p = 0.3$). Mosaic augmentation is disabled for the calc expert because it concatenates four images, disrupting the spatial density of calcification clusters and artificially increasing background complexity. Mosaic does not affect mass detection and is left at default for other models.

The *universal baseline* is a YOLO11l trained on the full canonical_4class partition (2190 images, four classes) with the identical training configuration. This represents the conventional single-model approach that jointly optimises all four classes.

Expert validation performance (mean over three seeds) is summarised in Table 2.

4.4 Domain-Split MoE Inference

At inference, the mass expert and calc expert both process every test image independently, as illustrated in Fig. 1. Their prediction sets are concatenated after applying the class offset: mass expert predictions retain local class indices 0 (BM) and 1 (MM); calc expert predictions are remapped to global classes 2 (BC) and 3 (MC) by adding the fixed offset +2. Because the two class ranges are disjoint, no ambiguity arises in the merged output. Standard confidence-threshold filtering ($\tau = 0.25$) and non-maximum suppression (IoU threshold 0.45) are then applied to the merged prediction set.

Table 2. Expert detector validation mAP50 (mean, three seeds, evaluated on expert-specific val partitions of 26/17 images respectively).

Expert	mAP50	Per-class AP50				Epochs
Mass Expert	0.499	BM 0.264	MM 0.734			∼107
Calc Expert	0.167	BC 0.189	MC 0.145			∼127
Universal	0.062	BM 0.066	MM 0.124	BC 0.014	MC 0.024	∼90

DS-MoE requires no routing decision and therefore avoids routing errors entirely. Its computational cost is that both expert networks run at every inference step, doubling FLOPs relative to the universal baseline.

4.5 Gated MoE with Router

The Gated MoE introduces a lightweight router to modulate expert confidence scores before merging. The router is a YOLO11 classification model trained on canonical_router (760 training images, binary labels: mass-dominant vs. calcification-dominant) for 50 epochs. It achieves mean top-1 accuracy of 0.712 across three seeds (individual seeds: 0.710, 0.710, 0.716).

Let $s_{m,i}$ denote the confidence of the i-th prediction from the mass expert and $p_m \in [0, 1]$ the router probability for the mass class. Modulated scores are computed as:

$$\tilde{s}_{m,i} = p_m \cdot s_{m,i}, \quad \tilde{s}_{c,j} = (1 - p_m) \cdot s_{c,j}, \tag{1}$$

where $s_{c,j}$ is the j-th calc expert prediction score. The merged set $\{\tilde{s}_{m,i}\} \cup \{\tilde{s}_{c,j}\}$ is then processed identically to DS-MoE with the same confidence threshold and NMS parameters. In principle, gating suppresses predictions from the non-dominant expert for each image; in practice, as Section 5.2 shows, the 71.2% router accuracy is insufficient to provide reliable modulation across the full test distribution.

4.6 Evaluation Protocol

Two metrics are reported throughout. Mean average precision at IoU threshold 0.50 (mAP50) is computed using the standard 101-point interpolated precision-recall curve aggregated over all four classes. FROC sensitivity at two false positives per image (S@2) is computed with an IoU matching threshold of 0.30, which reflects the lower localisation precision acceptable in clinical screening CAD and matches the evaluation convention of the Digital Mammography DREAM Challenge. Per-class variants of both metrics are computed by treating each class separately. All evaluations run on the fixed 458-image CBIS-DDSM test set with the 518 ground-truth bounding boxes described in Section 3.

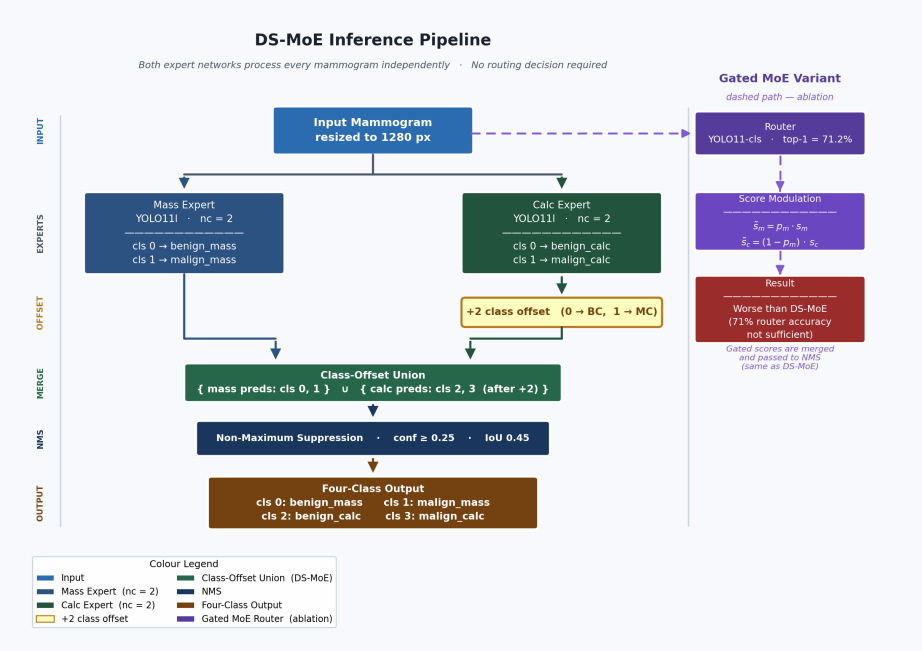


Fig. 1. DS-MoE inference pipeline. Both expert detectors run on every input mammogram independently. Calc expert predictions are remapped to global classes 2–3 via a fixed +2 class offset. Predictions are concatenated and filtered with NMS. The Gated MoE variant (dashed path) modulates confidence scores using router probabilities $\tilde{s}_{m,i} = p_m \cdot s_{m,i}$ before merging.

5 Experiments

5.1 Implementation Details

All models are trained using the Ultralytics YOLO11 framework [8] on NVIDIA T4 GPUs (14.9 GB VRAM) via the Kaggle platform. Mixed precision (AMP/FP16) is enabled throughout. Each seed produces independently initialised model weights; no weight sharing occurs across seeds or across expert/universal models within a seed. Total training time is approximately 0.5 hours per expert and 2.5 hours for the universal model per seed.

5.2 Main Results

Table 3 reports the main comparison. DS-MoE achieves mean mAP50 of 0.0649, a 13.9% improvement over Universal (0.0570), driven entirely by mass class gains. Absolute mAP50 values are modest across all models by design: strict patient-level evaluation removes the label leakage that inflates image-level-split benchmarks by 30–50% [4], so these numbers should be read as controlled relative

Table 3. Main comparison on the 458-image CBIS-DDSM test set (mean ± std, three seeds 42/1/123). All models use identical patient-level splits; direct comparison with prior work using image-level splits is not meaningful (see Sec. 2).

Model	mAP50	mAP50-95	FROC S@2
Universal	0.0570 ± 0.0014	0.0243 ± 0.0003	0.2021 ± 0.0071
DS-MoE	$\mathbf{0.0649 \pm 0.0114}$	$\mathbf{0.0300 \pm 0.0049}$	$\mathbf{0.2027 \pm 0.0114}$
Gated MoE	0.0589 ± 0.0075	0.0263 ± 0.0034	0.1950 ± 0.0150
Δ DS-MoE vs Universal: mAP50 +13.9%; FROC S@2 +0.3%			

Table 4. Per-class mAP50 (mean over three seeds). Bold = best per column. DS-MoE gains are concentrated in mass classes.

Model	Ben. Mass	Mal. Mass	Ben. Calc	Mal. Calc
Universal	0.0664	0.1236	**0.0144**	0.0237
DS-MoE	**0.0843**	**0.1421**	0.0099	0.0234
Gated MoE	0.0802	0.1133	0.0082	**0.0338**
Δ DS-MoE	+27.0%	+15.0%	−31.3%	−1.3%

comparisons under identical conditions rather than against leaked-split literature. Aggregate FROC S@2 is 0.2027 for DS-MoE versus 0.2021 for Universal—a difference of 0.0006 far below any meaningful clinical threshold, confirming no aggregate regression in detection completeness (Fig. 2). Gated MoE scores below DS-MoE on both metrics: with 71.2% router accuracy, incorrectly classified images have their correct expert down-weighted, causing true positive boxes to fall below the confidence threshold after gating.

5.3 Per-Class Analysis

Tables 4 and 5 decompose performance into per-class contributions, visualised in Fig. 3. DS-MoE wins on mAP50 for mass classes (benign_mass +27.0%, malignant_mass +15.0%) while Universal retains the advantage on per-class FROC S@2 across all four classes (Section 7 explains why this divergence arises). Both calcification mAP50 classes are slightly worse under DS-MoE, confirming that the calc expert introduces additional false positives without proportional true positive gain—consistent with its lower validation mAP50 of 0.167 versus 0.499 for the mass expert.

Notably, Gated MoE achieves a 42.6% improvement on malignant_calc mAP50 (0.0338 vs 0.0237), the only class where gating outperforms both baseline models. Malignant calcification is arguably the highest-cost false-negative class in

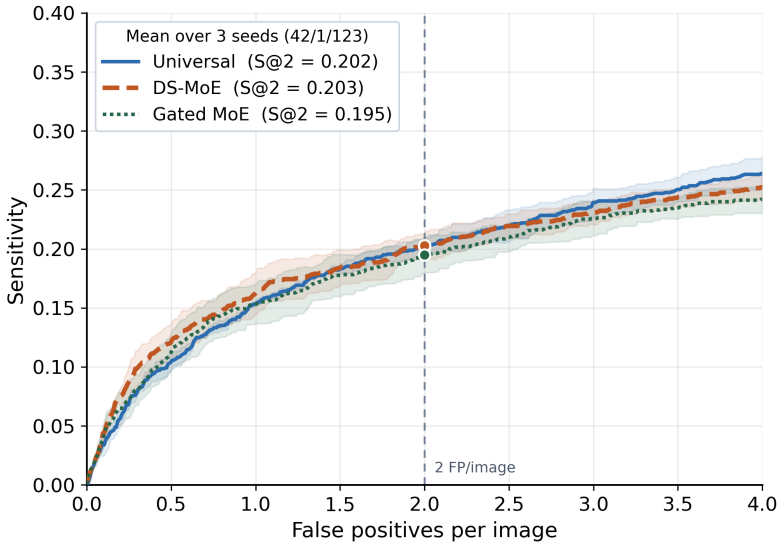


Fig. 2. FROC curves on the 458-image CBIS-DDSM test set (class-aware matching, IoU matching threshold = 0.30; mean over three seeds, shaded band = ±1 standard deviation). At the clinically relevant operating point of two false positives per image (dashed line), the three models are statistically indistinguishable (S@2 = 0.202, 0.203, and 0.195 for Universal, DS-MoE, and Gated MoE respectively), confirming that domain splitting introduces no aggregate sensitivity regression. Absolute sensitivity is conservative by design: patient-level leakage-free evaluation yields lower numbers than the image-level splits common in prior work.

screening; confidence modulation selectively suppresses low-confidence calc false positives while preserving the highest-confidence true positives.

5.4 Per-Seed Results

Table 6 shows per-seed results. DS-MoE wins on mAP50 for seeds 1 and 123 but not seed 42, where the mass expert trained on that seed had lower validation mAP50 during training. This seed-level variance underscores the importance of multi-seed evaluation and motivates future work with a larger seed pool.

5.5 Ablation: CLAHE Preprocessing

Table 7 reports the effect of Contrast Limited Adaptive Histogram Equalisation (CLAHE) applied as preprocessing to the calcification expert. CLAHE reduces mAP50 by 38.6% (0.070 to 0.043) and causes early stopping at epoch 55 versus 121 for the baseline, indicating that CLAHE prevents model convergence. The explanation is that CBIS-DDSM images are already preprocessed JPEGs;

Table 5. Per-class FROC S@2 (mean over three seeds, IoU matching threshold = 0.30). Universal achieves higher per-class sensitivity; the aggregate tie in Table 3 is explained by FP inflation analysis in Section 7.

Model	Ben. Mass	Mal. Mass	Ben. Calc	Mal. Calc
Universal	**0.4331**	**0.5377**	**0.2215**	**0.2937**
DS-MoE	0.3696	0.4340	0.1972	0.2574
Gated MoE	0.3673	0.4308	0.1728	0.2706

Table 6. Per-seed test results (458-image test set, IoU matching threshold = 0.30 for FROC).

Seed	Model	mAP50	mAP50-95	FROC S@2
42	Universal	0.0556	0.0239	0.2027
	DS-MoE	0.0519	0.0245	0.1873
	Gated MoE	0.0484	0.0215	0.1737
1	Universal	0.0590	0.0245	0.2104
	DS-MoE	0.0798	0.0364	0.2066
	Gated MoE	0.0660	0.0290	0.2046
123	Universal	0.0564	0.0244	0.1931
	DS-MoE	0.0631	0.0292	0.2143
	Gated MoE	0.0621	0.0284	0.2066

CLAHE distorts the subtle pixel-level intensity patterns that YOLO uses to identify calcification specks and disrupts the pretrained COCO feature maps. This negative result is directly useful to practitioners: CLAHE, while widely recommended in medical imaging preprocessing pipelines, is demonstrably harmful for YOLO-based calcification detection on CBIS-DDSM.

6 External Validation

6.1 INbreast Zero-Shot Transfer

INbreast [5] is a full-field digital mammography dataset acquired at a separate institution with different equipment. We apply CBIS-DDSM-trained models directly without fine-tuning. Calcification evaluation is excluded: INbreast annotates individual calcification specks as point locations rather than cluster bounding boxes, making cluster-level IoU ≥ 0.30 structurally unmeasurable. We report mass S@2 only on 116 ground-truth mass boxes from 107 images.

Table 8 shows DS-MoE achieves mean mass S@2 of 0.8305 ± 0.0400 versus 0.7443 ± 0.0226 for Universal, a mean improvement of $+0.0862$ (11.6%).

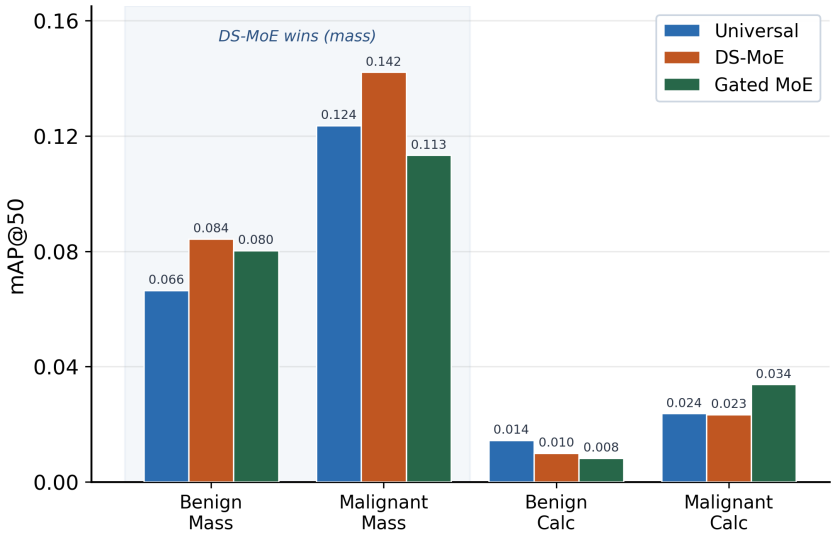


Fig. 3. Per-class mAP50 (mean over three seeds). Domain splitting concentrates its gains on the two mass classes (shaded), where DS-MoE clearly outperforms the universal baseline (+27.0% and +15.0%), while calcification performance remains uniformly low across all models, identifying the calc expert as the system bottleneck.

Seeds 1 and 123 show clear advantages of +0.11 and +0.15. Seed 42 produces an exact tie at 0.7759, where both models detect exactly 90 of the 116 boxes at the 2 FP/image operating point—a FROC curve threshold coincidence, not model equivalence. The mass specialisation advantage persists in zero-shot transfer across acquisition modality (screen-film to digital), confirming the benefit is not dataset-specific.

6.2 MIAS

The mini-MIAS dataset contains 322 images with 119 ground-truth boxes (53 benign mass, 39 malignant mass, 15 benign calc, 12 malignant calc). DS-MoE achieves mAP50 of 0.0917 ± 0.0063 versus 0.0880 ± 0.0330 for Universal (+4.2%, consistent across all three seeds). Overall FROC S@2 favours Universal (0.4314 ± 0.0241 vs 0.3922 ± 0.0317); however, only 27 calcification ground-truth boxes exist across 322 images, making FROC unreliable at this operating point. The consistent mAP50 improvement confirms that mass specialisation generalises to MIAS, a different screen-film dataset.

6.3 CMMD

The Chinese Mammography Database (CMMD) provides image-level cancer labels without bounding box annotations. We therefore measure image-level detection sensitivity: a test image is counted as detected if at least one prediction

Table 7. Effect of CLAHE preprocessing on the calcification expert (seed 42, evaluated on the expert val partition; training converged at 121 epochs without CLAHE versus 55 with CLAHE).

Config	**mAP50**	**Ben. Calc**	**Mal. Calc**
No CLAHE (ours)	**0.070**	**0.060**	**0.080**
CLAHE (c=2.0, 8×8)	0.043	0.023	0.063
Δ	−38.6%	−61.7%	−21.3%

Table 8. Zero-shot transfer to INbreast: mass-only FROC S@2 (116 GT boxes, 107 images). Calc excluded: INbreast uses point annotations incompatible with cluster-level IoU ≥ 0.30.

Seed	**Universal**	**DS-MoE**	Δ
42	0.7759	0.7759	+0.0000
1	0.7328	0.8448	+0.1121
123	0.7241	0.8707	+0.1466
Mean ± Std	0.7443 ± 0.0226	$\mathbf{0.8305 \pm 0.0400}$	+0.0862

box is produced above the confidence threshold. DS-MoE achieves image-level sensitivity of 0.1548 ± 0.0287 versus 0.0020 ± 0.0014 for Universal, a 77-fold improvement. The near-zero Universal sensitivity reflects the difficulty of the CBIS-DDSM-to-CMMD domain shift (screen-film to Chinese digital screening); DS-MoE mass specialisation substantially reduces this gap, demonstrating broader cross-scanner generalisation. Note that image-level sensitivity is a proxy metric in the absence of bounding box annotations; localisation accuracy cannot be assessed on CMMD.

7 Discussion

The central finding is that domain specialisation substantially improves mass detection precision (mAP50) while leaving calcification performance largely unchanged, and that the calcification specialist is the limiting component of the current system. At a 1280-pixel input resolution, a microcalcification cluster spanning 5–15 pixels in the original 3000 × 4000 pixel acquisition reduces to roughly 2–5 pixels—below the reliable receptive field of standard convolutional anchors. Training on a class-focused subset does not change the pixel budget available to represent each cluster.

Per-class FROC regression. Despite the aggregate FROC S@2 tie, Table 5 shows that Universal outperforms DS-MoE on all four per-class FROC metrics, with mass class losses of 14.7% and 19.3%. This apparent contradiction is explained by the inference design: running both expert networks on every image

doubles the total prediction count before NMS. For mass-dominant images, the calc expert contributes spurious false-positive boxes; these cross-domain FPs inflate the per-image false-positive count for the mass class, shifting the per-class FROC operating curve left and reducing the sensitivity achievable at the 2 FP/image threshold. The aggregate FROC tie arises because the FP inflation from mass-dominant and calc-dominant images approximately cancels across the mixed test set. This cross-expert contamination is the primary limitation of the unconditional "run all experts" strategy and motivates confidence gating as a direction—even though the current 71.2% router accuracy is insufficient to recover the loss.

Calcification bottleneck. We investigated native-resolution tiling (1024-pixel tiles with stride-4 P2 head) as an alternative. This approach plateaued at validation mAP50 of approximately 0.07 because tiling discards global tissue context that is important for calcification characterisation. Cluster boundaries are partially defined by their relationship to surrounding parenchyma, and short tiles with insufficient overlap lose this context. A useful calc expert likely requires large-overlap tiling with SAHI [9] at inference, or a dual-branch architecture combining high-resolution local crops with a global context stream.

External validity. A zero-shot experiment on VinDr-Mammo [13] showed Universal outperforming DS-MoE; we report this to avoid publication bias and attribute the finding to a domain gap specific to Vietnamese digital screening mammography—a different acquisition modality from CBIS-DDSM screen-film.

Limitations. Three seeds provide only a pilot variance characterisation; formal statistical claims require ≥ 10 seeds. Router accuracy of 71.2% is insufficient for reliable confidence gating. The calc expert was trained with full-image inference only. CMMD evaluation is image-level sensitivity only; localisation accuracy on that dataset cannot be verified.

8 Conclusion

We presented DS-MoE, a domain-split mixture-of-experts framework for four-class mammography lesion detection. Specialist YOLO11l detectors for mass and calcification subtypes are trained independently and merged at inference through a class-offset union. On CBIS-DDSM, DS-MoE achieves 13.9% higher mAP50 than a leakage-free universal baseline with mass class improvements of 27% and 15%. Aggregate FROC S@2 is clinically equivalent (0.2027 vs 0.2021); per-class FROC regression in DS-MoE is identified as a consequence of cross-domain false-positive inflation from the unconditional dual-expert inference, not a fundamental sensitivity loss. Zero-shot validation on INbreast (+11.6% mass FROC), MIAS (+4.2% mAP50), and CMMD (77× sensitivity) confirms cross-scanner generalisation. The calcification expert is the identified bottleneck, pointing to resolution-aware tiling with SAHI and improved routing as the key future directions.

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